

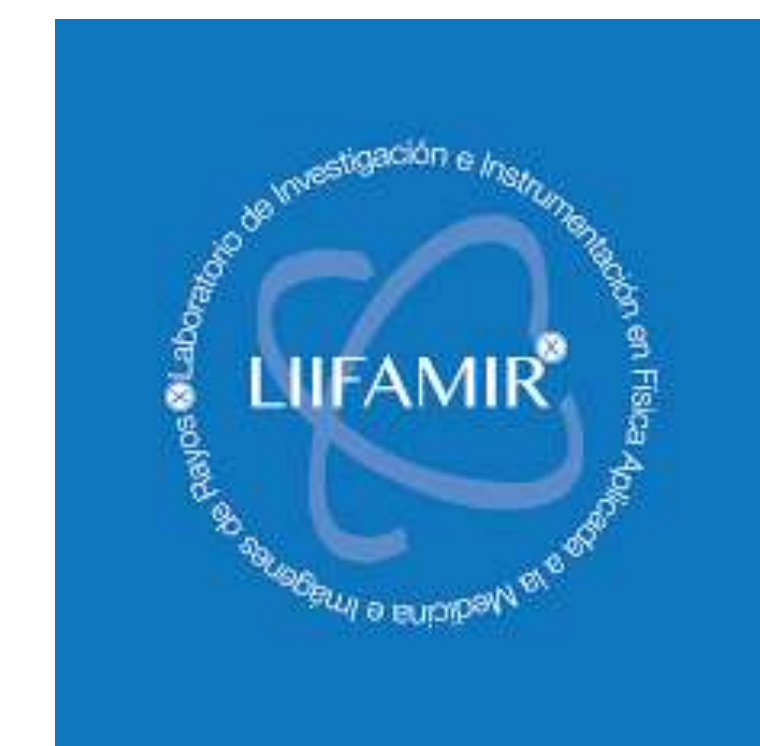
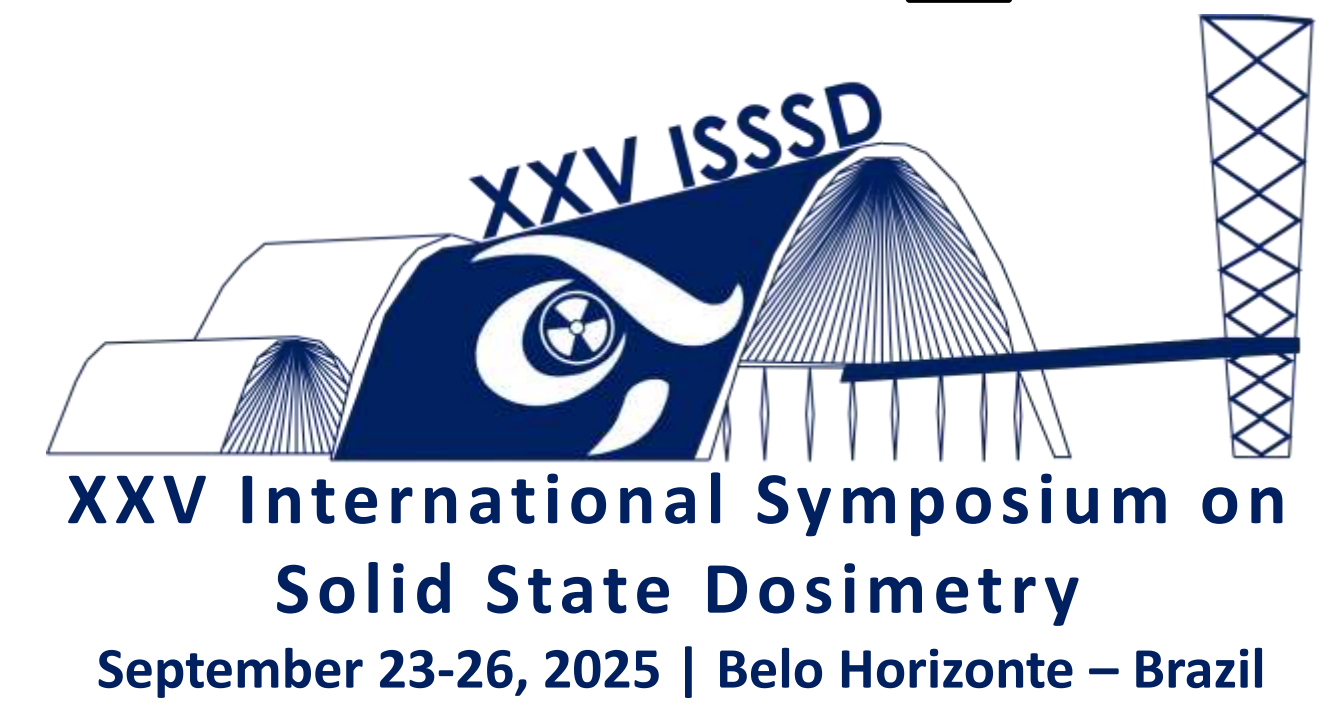
Preliminary experimental characterization of magnetic fields for dosimetric modelling with Monte Carlo methods in MR-Linac radiotherapy

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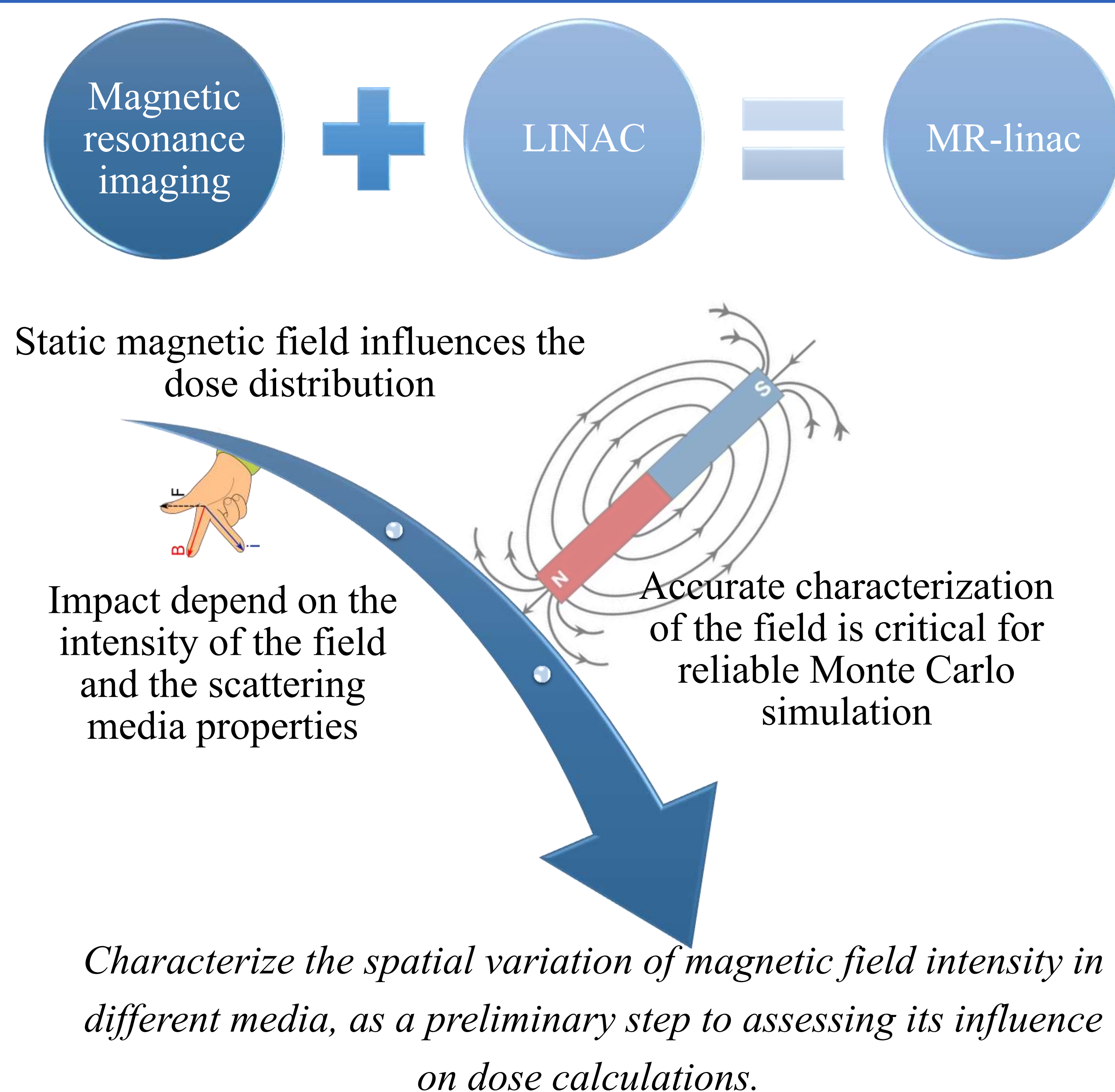
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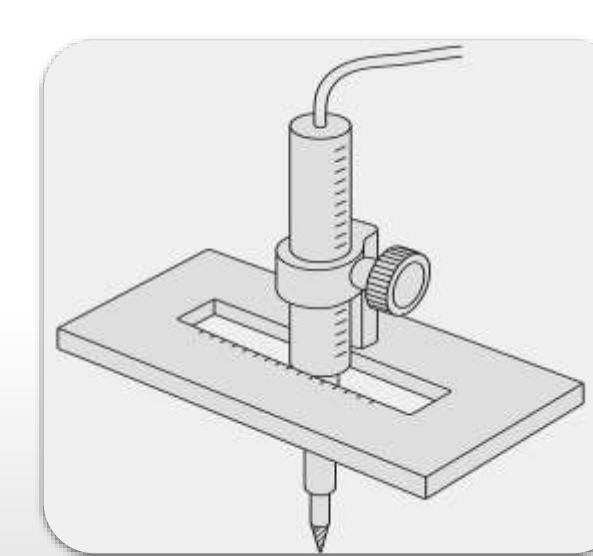
I. INTRODUCTION



II. MATERIALS and METHODS



Measurements were conducted with Varian permanent magnets (model 00-950002, up to 2 T) from NMR Laboratory



Vertical sensor positioning system was developed for SJ200 gaussmeter due to limited magnets polar faces access; Measurements were performed at two fixed positions: 1.88 T and 1.90 T.



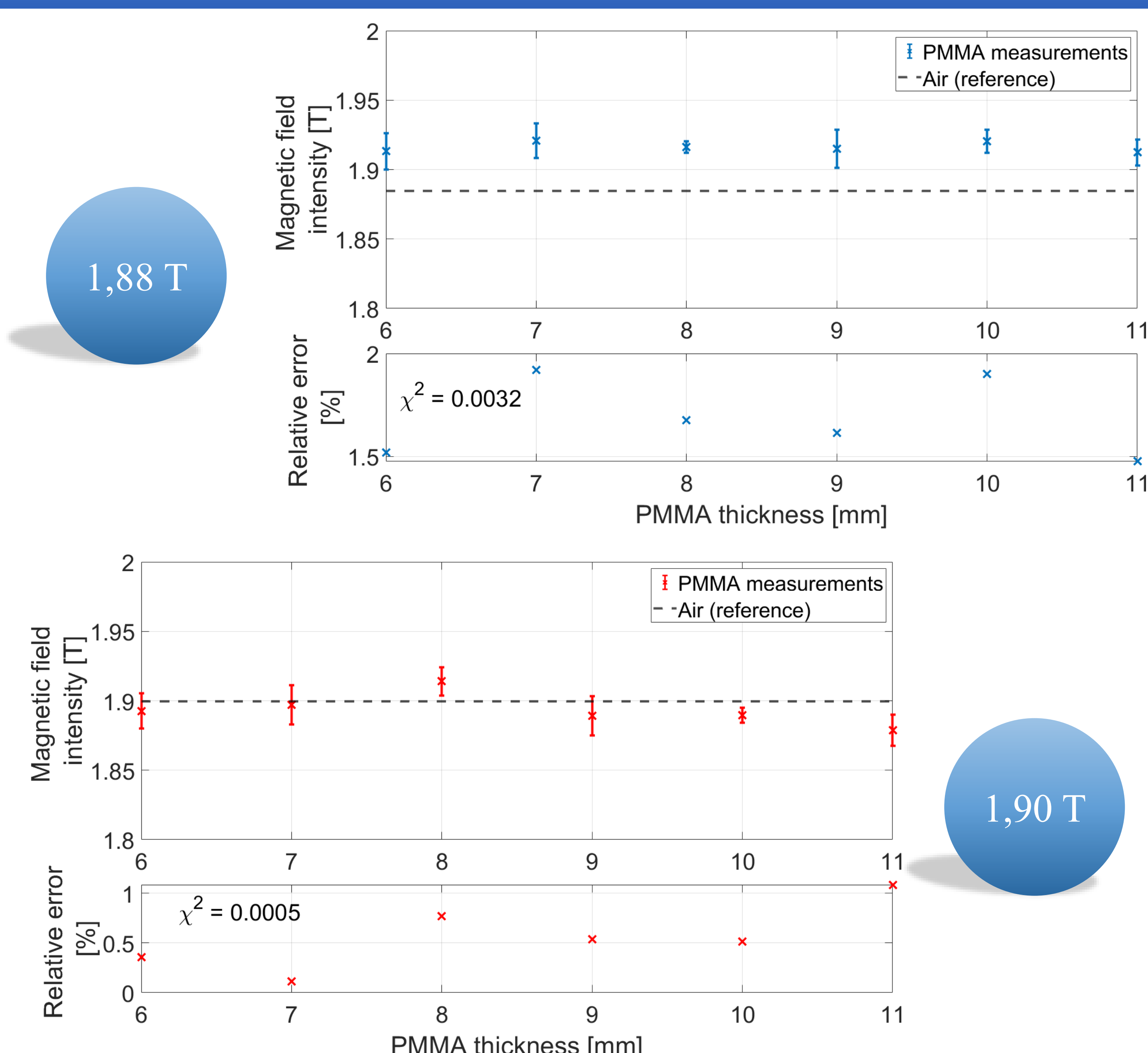
Air vs PMMA cap measurements (6-11 mm thickness) were performed with complete sensor coverage; PMMA was used as tissue-water-equivalent standard

$$\mu = \frac{1}{n} \sum x_i$$

$$\sigma_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

Field intensity was measured five times per thickness and averaged with uncertainties; Mean values were plotted against thickness to analyze functional dependence.

III. RESULTS



IV. CONCLUSIONS

No statistically significant variations between air reference and PMMA-covered sensor configurations

Relative error ($\leq 3\%$) remained within the experimental uncertainty ($\pm 2\%$)

$\chi^2 \ll 1$, confirming a strong correlation between air and PMMA measurements

Strong dependence on measurement location highlights the need for

Systematic field mapping across different intensities

Improved positioning systems

Comprehensive characterization before establishing definitive correction protocols

Preliminary results suggest that field corrections may not be required for MR-Linac dosimetry simulations, though systematic validation across field configurations remains necessary for robust protocols.

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